Astronomical Telescope "Hadley" - an easy assembly, high performance Newtonian

Maff

VIEW IN BROWSER

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Summary

This telescope assembles with only prints, screws and glue - it is light and easy to use, and total project cost < $150

32.53 hrs	10 pcs	0.45 mm 0.25 mm 0.48 mm	0.60 mm	PLA	831 g	Prusa MK3/S/S+

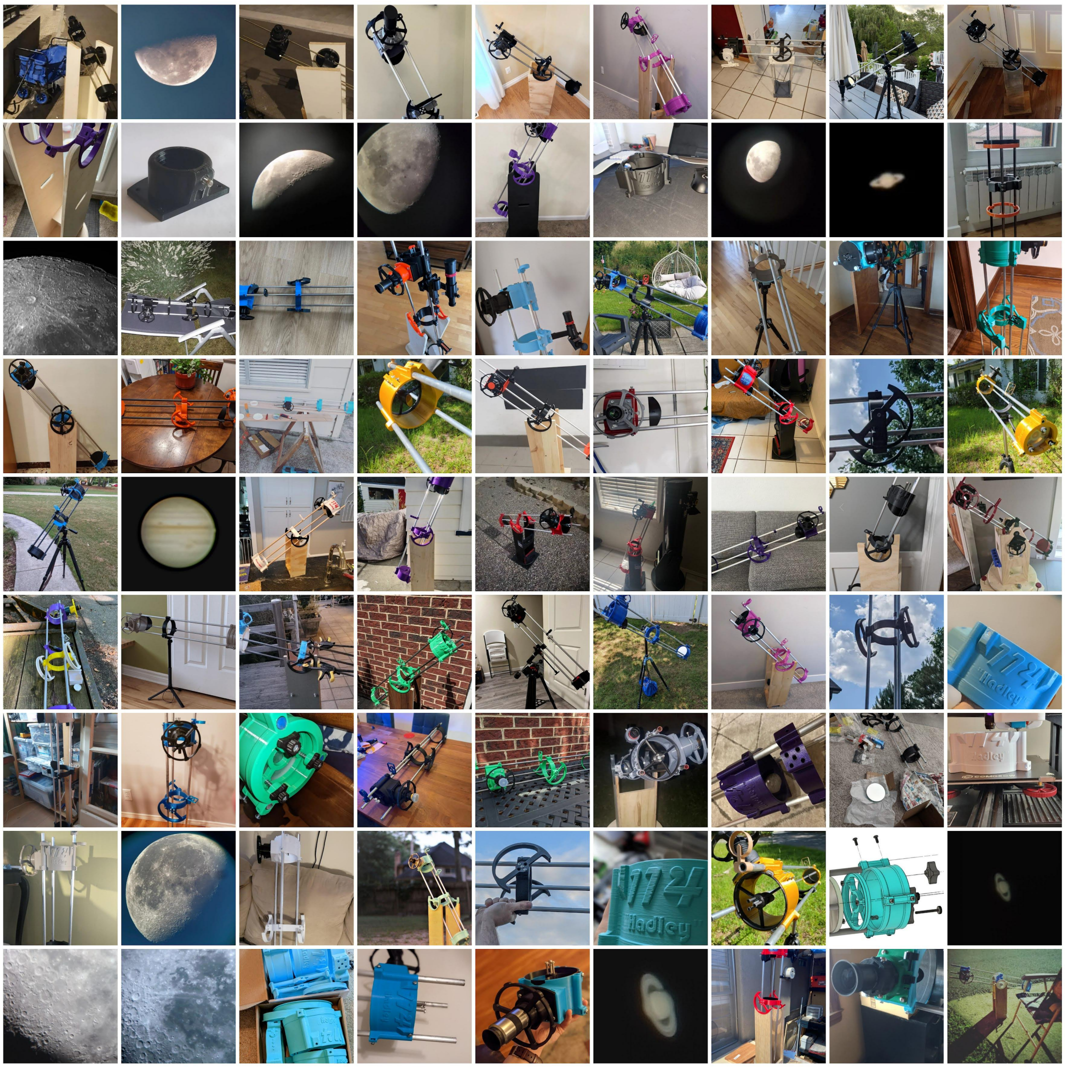
Learning > Physics & Astronomy

Tags: space education stem science moon telescope astronomy astrophotography physics photography saturn planets newtonian reflector birdwatching dobsonian

You can now purchase screw/mirror kits at Kissner Optik.
Official Discord server here.
New instruction manual is here! See “other files.”
I recommend the Hill Mount, Marci Mount or EMT remix over the one featured here. Wooden ones still are the best, but these pipe mounts are excellent, especially paired with a wooden friction bearing.
Metric users: click here, and see other remixes.
Lastly, never point one of these at the sun - it can melt your eyes.

Featured in Sky and Telescope magazine, this PetaPixel Article, Hackaday and others.

The initial wave of support/adoption has been overwhelming - I've confirmed hundreds of separate Hadley's out in the wild, and napkin math suggests roughly 3000 more than that. Thank you.



Early adopters: Please check in from time to time. I am revising this based on user feedback. Immediate plans include proper non-wooden mount options.

Meet Hadley.

Hadley is a very performant telescope that - after printing - should be as easy as IKEA furniture to assemble. Just screws and glue.
Building a telescope is the best way to learn how one works.

The mission here is to make an attractive alternative to the shoddy, hard to use "hobby-killer" scopes in the $100-200 range. It still needs a mount (basic woodworking or a beanbag, or even a sturdy clamping tripod for now), but it performs intuitively and flawlessly. "Hadley" is a 114/900mm reflector with a spherical primary, effectively a perfect parabola at this size.

This functions equally well as a terrestrial scope (great for bird-watching) or a slow (f8) zoom lens for a camera - but it really shines in its primary purpose, which is for visual and photographic astronomy on planets and the moon. It is my hope that by releasing this, more people will find an open door to planetary observation and entry level astrophotography.

CHANGELOGS:

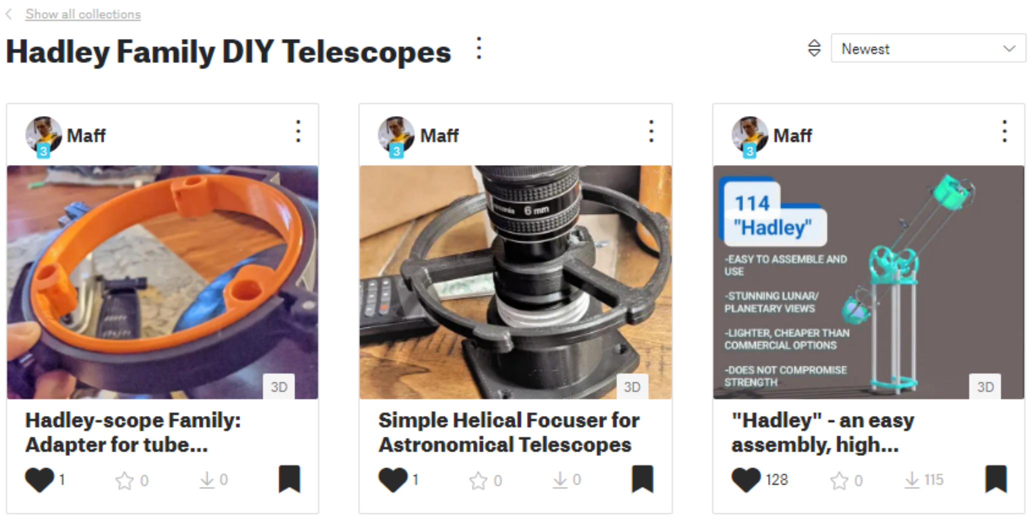
- Updated UTA to have better pockets
- Uploaded a new LTA with the knob array rotated. This is for clearance with larger knobs.
- Took an axe to the directions here. Please see the attached PDF instead.
- **Added instruction manual v0**
- Added better knobs, other tweaks

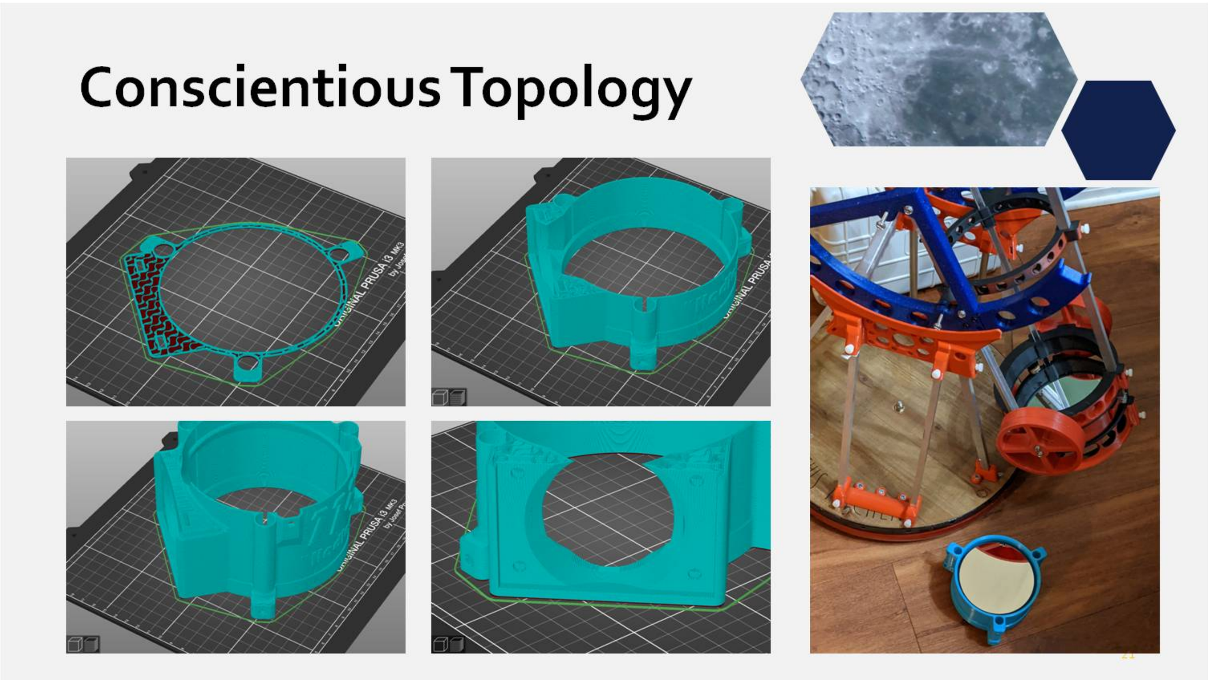
  2023
- Endorsed wigglemore's version as **the new official metric version**
- **Added several revised spiders**
- Launched a small shop for a few people wanting kits

  August
- Added STEP files
- Began support for a metric version
- Added a nice printed rocker to the linked family of mods.
- Added a synta/dovetail shoe as an add-on. Use this to attach a finderscope or red dot sight!

  July
- Added a light baffle as an add-on
- Scaled up the knobs for better clearance
- added a printed baffle. Install upside-down, opposite focuser under the UTA.

- Added several “mods” for Hadley, as well as some one-off parts such as the focuser - for other DIY astronomers.
- Created a collection, where I will store these to avoid cluttering up the main page.
- Added a means to mount equatorially.
- Added a “TEST PRINT” file for making sure your selected screws and rods (M4 for metric users) fit / that your printer is dialed in enough.
- There is now an amazon listing for mirrors! A friend grabbed three from there, it seems to check out.



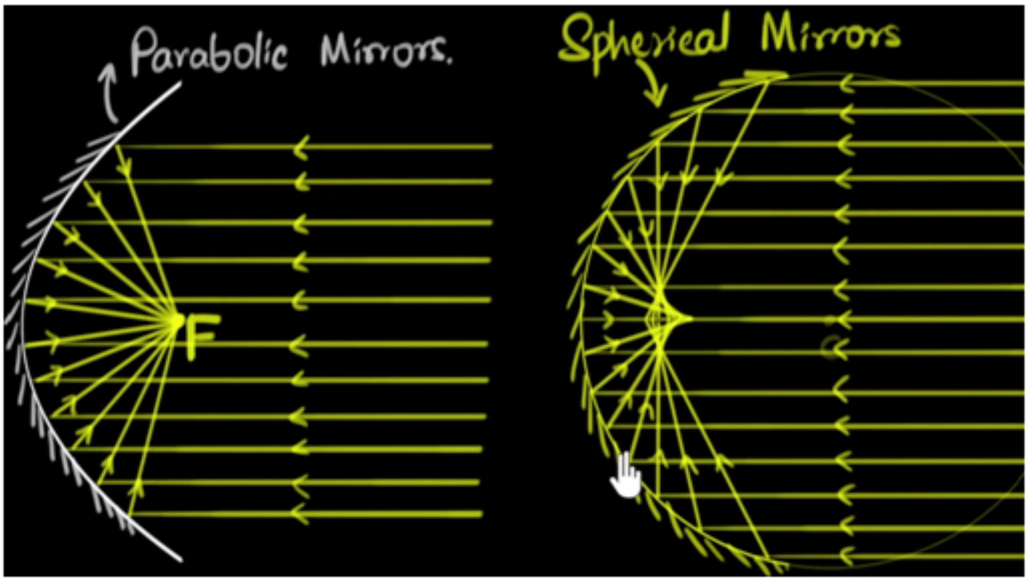


Design: Conscientious Topology, or “Prints Without Supports”

This is the second iteration on this idea after my first telescope -- a large red reflector (152/1300mm). My first project was personal and hard to build; The "Hadley" arose as a quest for universal design/accessibility, practicality and replicability.

Each part was designed bottom to top with FDM in mind, and underwent a lot of testing to ensure it works well. Every STL comes in the ideal orientation, and for everything but the focuser (which is reinforced by the backspan of the eyepiece barrel) - the parts are oriented not only to **print support-free,** but also so that there's minimal shearing in the plane of the layers - everything is in its strongest orientation for its job.

It only uses one type of screw thread (in two or three lengths), one type of nut, and simple/cheap mirrors - the best compromise in cost/performance.



Khan Academy

APERTURE	TEXEREAU MINIMUM F/RATIO
3 inch	f/6.4
4 inch	f/7.1
6 inch	f/8.1
8 inch	f/8.9
10 inch	f/9.6
12 inch	f/10.2

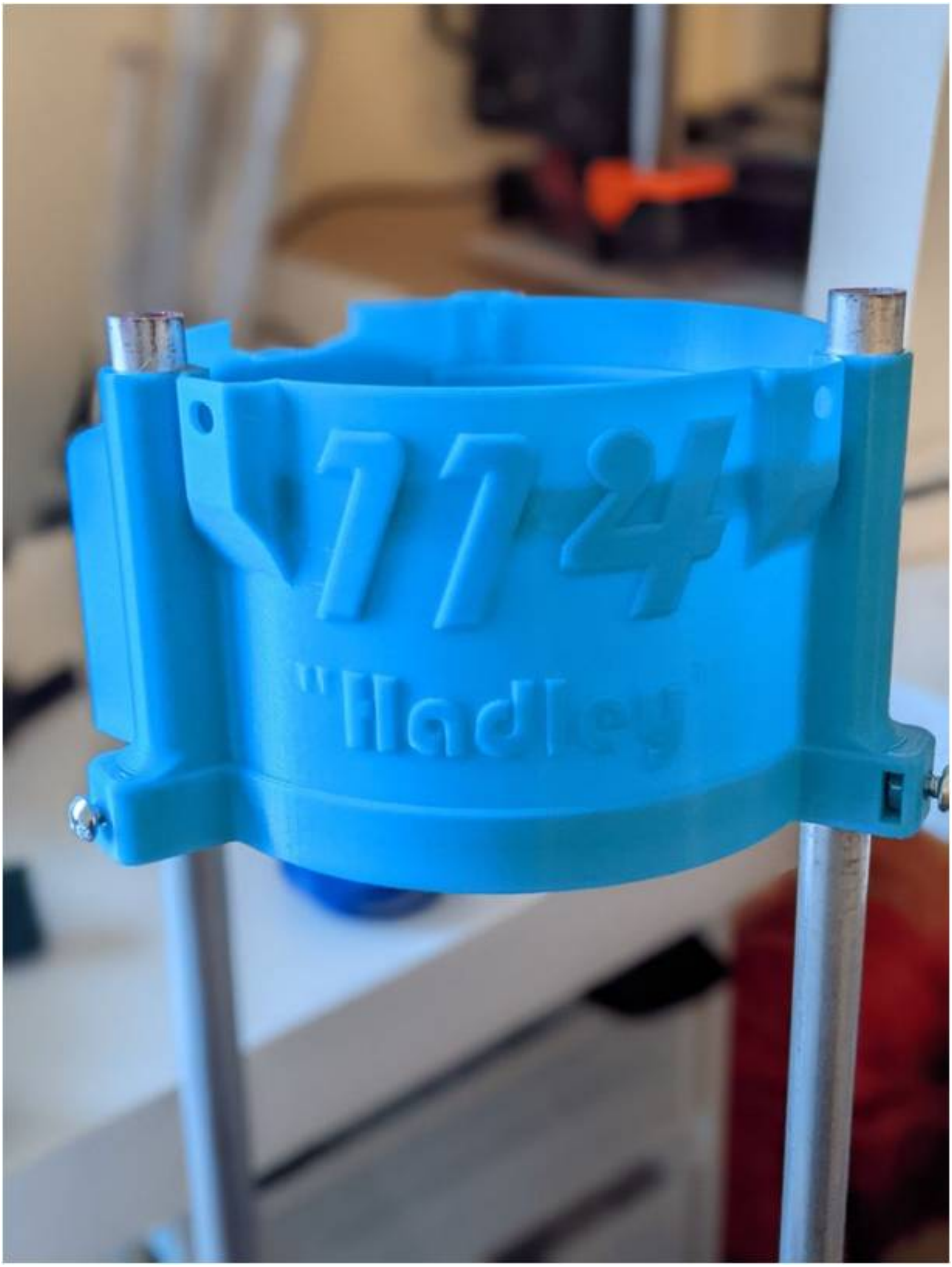
Texereau

The choice of mirror

It happens that parabolic mirrors are prohibitively expensive. But in certain focal ratios/sizes, a spherical mirror is a close enough stand in. And this is big, because a mirror with a spherical cross section is (comparably) trivial to make to near-molecule precision... considering.

As it happens, many commercial scopes make a similar compromise, although most of these are well outside the "good" range. Exercise caution when buying a commercial telescope in the 100-150mm range.

At time of posting, these mirror sets (114/900mm spherical primary and elliptical secondary pair) can be found for about $20 on ebay or aliexpress. This is prone to fluctuation.



3D - printing a telescope

This is a fairly involved project -- it takes around $100 in material (plus eyepieces, a good pair might total $50) but extremely rewarding.

Printing

Please read the descriptions on the files/the attached README. **There are many redundant items.**

NEED-TO-KNOWS

NEVER POINT IT AT THE SUN
Seriously - remember frying ants in a magnifying glass? Well this is a really big magnifying glass. For more advanced users looking to do solar observing with proper filters - be aware the open design of Hadley will focus sunlight outside of the optical path as well, so this really needs a full baffle (a cardstock tube, even).

AVOID HEAT
This is true of any print in PLA - if it reaches 110F, say in a hot car, the prints will deform. ABS/PETG makes can worry less, though I have had PETG (slightly) deform in the Texas heat.

CLEANING THE MIRRORS?
Firstly: Never use a towel, rag etc. on your mirrors.
These are “first-surface” optics - completely different from mirrors and lenses you may be used to. There is no glass protecting the optics, indeed they are just a molecules-thick layer of aluminum on an impossibly smooth glass surface. Both of these are incredibly easy to damage. Fingerprints/ dust won't hurt the view as much as you think. If the mirrors are truly dirty enough to hurt the views, look for detailed guides that mention cleaning with your fingertips.

Eyepieces are a different story - handle with care, but you can clean them carefully using lens wipes.

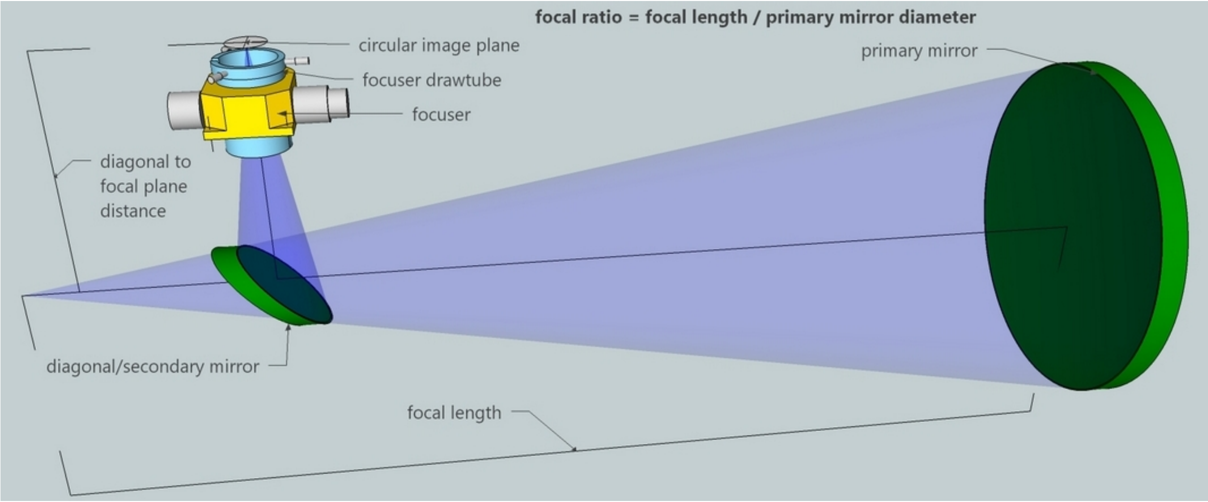
Assembly Instructions/Bill of Materials

BILL OF MATERIALS

A: Get a full screw/mirror/eyepiece kit from me at Kissner Optik!

B: Metric users: Scroll through the remixes, there is now full support!
C: Imperial unit users:

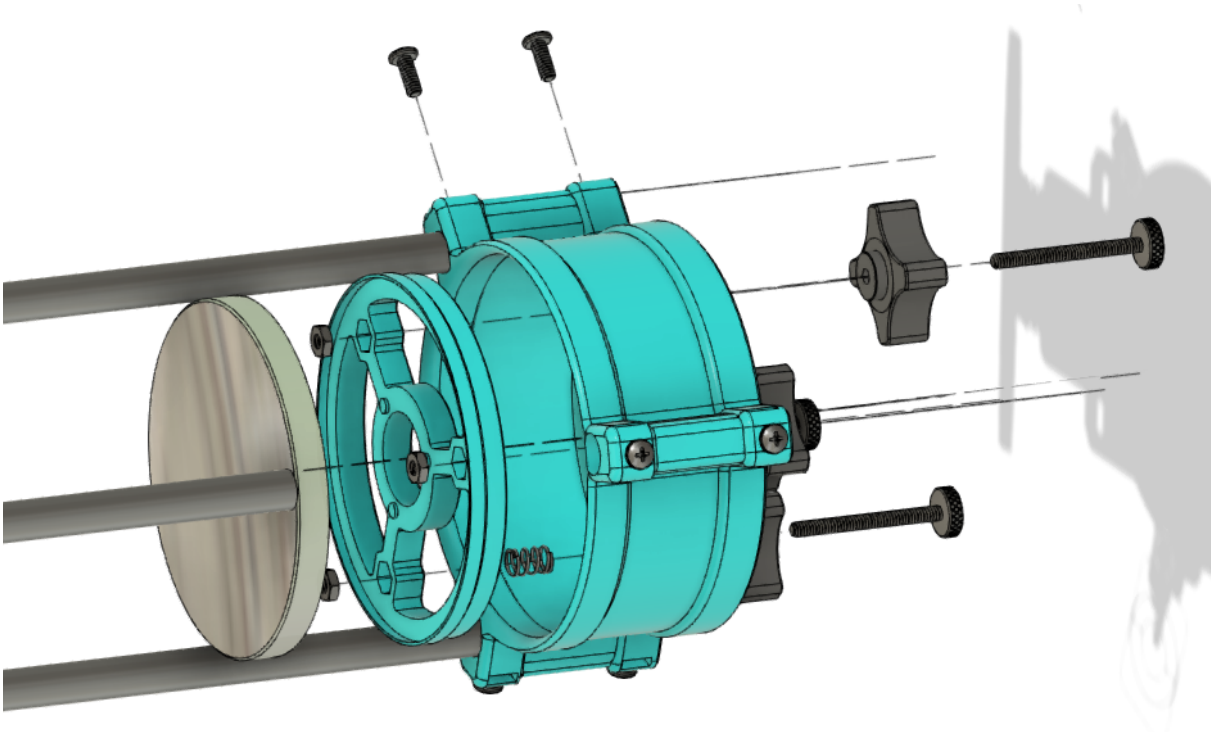
See the materials guide or full assembly manual. You ultimately need about $80 in springs, screws, nuts, optics and metal rods - all very easy to source online/ in stores.



Mel Bartels, https://www.bbastrodesigns.com

Assembly

Assembly is straightforward and flexible – **read the attached manual.**



FAQ

What can I see with Hadley?
The moon and its craters, seas and rilles in incredible detail. Rich planetary detail - the rings of saturn (as shown); storms on Jupiter, dust clouds and canyons on Mars. A few of the brightest deep sky objects - nebulae, certain galaxies, and rich clusters of stars - depending on how dark your skies are.

Do I need to center-dot my mirror?
In my opinion, collimation can be achieved with a cap without center marking the mirror. F/8 is slow enough that collimation tolerances are loose.

What if I use metric/ can't find imperial screws?
For now, there are remixes available and notes under said remixes. Official support for m4/m5 hardware is being actively developed.

Which spider do I use?
That's up to you – look for the picture at the top which discusses diffraction spikes. You might notice all images from the famous "Hubble Space Telescope" have four pointed stars, and JWST's images all have six points (ignoring all the other diffraction artifacts). This is due to the secondary mirror support – and with Hadley, you get to pick which one you have!

What parts need a captive nut?
Just the spider. Heat-serts and other remix versions are on the way. If you're printing your own thumb-nuts, those need a captive nut as well.

Why did I start with imperial? My prototype was mostly M4 - but it was such a hodgepodge of differing lengths, threads etc. I realized if I released it, the BoM would be prohibitively complicated and expensive, like many other 3DP telescope designs out there. I wanted to avoid that pitfall. I wanted this new version to be doable without ordering anything (except the mirror). So to support that - I only used what I could find in the stores. Sadly, metric machine screws are much harder to find in bulk here.
I am working to provide a modified spider to accommodate the taller M5 nuts.

Can I put this on an equatorial mount/use it for astrophotography?
Yes, although it's a little "slow." I provide modified files in my **collection** on this site. An interface for 6 inch tube rings is found **here.** If you want to make a Hadley Astrograph, try an f/4 mirror, heavier springs and a larger secondary (44mm).

Can I attach a finderscope?
Yes, same as above. I've provided an **interface** for a 6x30, and a dovetail shoe is now included. A TELRAD interface shouldn't be too hard to make, either.

Can you add x part?
Within reason, I want to make this as functional as possible. But my time is scarce as a grad student and job seeker so things are slow.

Does this scale up or down?
Yes, but so do the screw-holes and engineering challenges. I have someone trying to make an eight-inch Hadley, so we'll see how that goes.

Longer term, I will release 5 and 6 inch versions, as well as refractor versions.

Modifications, etc:
You can find extra parts, improvements, alternates in my collection **here**. Support these files as well!

Discord: Where can I get more support/collaborate?
I am trying out a discord server for this project.
Additionally, this Observational Astronomy server is a phenomenal space for space. Connect with other amateur astronomers, get advice, do observing challenges and more!

Message #resources

How can I support this?
Tips are always appreciated, as is support on **Printables**. I'm also (barely) on **Patreon,** where I host shelved projects I don't have time to provide support/documentation. Designing, testing and improving takes a lot of time, parts, and filament - and support gets me more of all three.

Support a project hundreds of hours in the making:

Patreon

Model files

Upper assembly

5 files

utar.stl

baffle.stl

OPTIONAL. This attaches upside-down, under the UTA opposite the focuser. Spray gloss black.

focusero.stl

Print at 0.2mm. Interior should be black.

focuseri.stl

Print at 0.2 or 0.25mm. Interior should be black.

secondaryholder.stl

Print solid. PRINT/PAINT BLACK.

Lower assembly

8 files

knob-p.stl

PAUSE-PRINT. Choose this or the Knob-B

knob-b.stl

Non-pause version of the knob. Chose this, tri-knob or Knob-P

tri_knob.stl

Non-pause version of the knob. Choose this or the above ones.

sightsncell.stl

fwdsight.stl

OPTIONAL: Sights split from SightsNCell

backsight.stl

OPTIONAL: Sights split from SightsNCell

cellonly.stl

OPTIONAL: Cell split from SightsNCell

lta-r.stl

Spiders -- PICK ONE -- PAUSE TO INSERT HEXNUTS

8 files

meme-spider-donotuse.stl

(‿)

spider_new_minimized.stl

New spider - pause once to insert nuts. (Experimental: Super thin vanes)

spider_hubble.stl

PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

spider_hubble_minimized.stl

PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

spider_jwst.stl

PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

spider_jwst_minimized.stl

PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

spider_illegal.stl

PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

spider_illegal_minimized.stl

PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

Middle Assembly

6 files

mount_interface_planets.stl

PICK ONE - Decide if you want planets or not

mount_interface.stl

PICK ONE - Decide if you want planets or not

mount_interface_wider_wood-rocker-only.stl

OPTIONAL: Wider mount interface, to give some extra clearance for a wooden rocker box

z_bearing2.stl

Print "sturdier" (30-50% cubic infill, more perimeters)

z_bearingb.stl

OPTIONAL - bearing file, but split to parts

z_bearinga.stl

OPTIONAL - bearing file, but split to parts

Extra-optional-SOURCE

6 files

test_print.stl

Use this to ensure the joints print well, and your chosen fittings work.

finderscope-interface.stl

OPTIONAL. Adds a dovetail interface for a finderscope. (Part by Rob Davidoff)

rockerbaseimperial.stl

rockertopimperial.stl

hadleysource.step

hadley_f360.f3z

Print files

Gcode files

10 files

spiderjwst_06n_045mm_pla_mk3s_1h43m.gcode

PLA 0.60 mm 0.45 mm 1.72 hrs 39 g Prusa MK3/S/S+

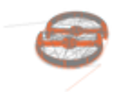
PICK ONE spider. See images for what these do, and remember this needs a pause/insert nut!

sightsncell_06n_045mm_pla_mk3s_1h39m.gcode

PLA 0.60 mm 0.45 mm 1.66 hrs 54 g Prusa MK3/S/S+

focusero_06n_025mm_pla_mk3s_3h23m.gcode

PLA 0.60 mm 0.25 mm 3.38 hrs 76 g Prusa MK3/S/S+

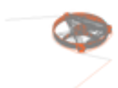


z_bearing2_06n_048mm_pla_mk3s_2h41m.gcode

PLA 0.60 mm 0.48 mm 2.68 hrs 102 g Prusa MK3/S/S+

mountinterface_planets_fine_06n_048mm_pla_mk3s_.gcode

PLA 0.60 mm 0.48 mm 5.80 hrs 144 g Prusa MK3/S/S+

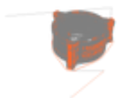


spider_illegal_06n_045mm_pla_mk3s_1h7m.gcode

PLA 0.60 mm 0.45 mm 1.12 hrs 35 g Prusa MK3/S/S+

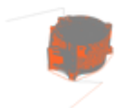
focuseri_06n_045mm_pla_mk3s_3h6m.gcode

PLA 0.60 mm 0.45 mm 3.11 hrs 77 g Prusa MK3/S/S+



ltafinallabe_06n_045mm_pla_mk3s_6h4m.gcode

PLA 0.60 mm 0.45 mm 6.07 hrs 146 g Prusa MK3/S/S+



uta-final_06n_045mm_pla_mk3s_8h23m.gcode

PLA 0.60 mm 0.45 mm 8.39 hrs 189 g Prusa MK3/S/S+

secondaryholder_06n_045mm_pla_mk3s_19m.gcode

PLA 0.60 mm 0.45 mm 0.32 hrs 8 g Prusa MK3/S/S+

Other files

readme.txt

spiders_old.zip

Contains the older spiders- these work, but users report trouble with the pause-prints.

hadley-instructions-v1.pdf

License